

Metazoan Parasite Infracommunities of the Freshwater Eel, *Mastacembelus armatus* Lacépède, 1800 from River Godavari, India

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ABSTRACT

Mastacembelus armatus is considered to be the delicacy of Southern India. Four hundred ninety four specimens of *M. armatus* collected from river Godavari from August 2005 to September 2007 were analyzed in order to study their metazoan parasite infracommunities. Twelve species of parasites were collected, 6 digenea, 2 cestodes, 1 monogenea, 1 copepoda, 1 nematoda and 1 acanthocephala; 78% of the fishes were parasitized by one or more than one metazoan, with a mean of 67 parasites/fish. The endoparasites represents 98.3% of the total parasites collected. The digenean *Tetracotyle* sp. and *Circumonchobothrium shindei* occupy the position of secondary species and the remaining were satellite species. Relationships between total body length of fish and both total parasite abundance and mean parasite species richness were observed. A new copepod species, *Neoergasilus indicus* is also encountered in the present investigation. The metazoan parasite infracommunities of *M. armatus* presented dominance of larval endoparasites; correlation of parasite burden, diversity and species richness with host total length; and no influence of host sex on parasitisation.

Key words: Parasite ecology, community structure, *Mastacembelus armatus*, *Tetracotyle* sp., *Circumonchobothrium shindei*

INTRODUCTION

Parasites are small players with crucial role in ecological theatres is well said by Marcogliese (2004) as he signified parasites as imperative tools for the effectual management of our natural resources by having an ecosystem based approach. Parasites can provide information on population structure, evolutionary hypotheses, environmental stressors, trophic interactions, biodiversity and climatic conditions. Parasites are metabolically dependent on their hosts mainly for their nutritional requirements (Marcogliese, 2004). However, it is an established fact that many species of parasites provoke pronounced or subtle effects on hosts affecting their behaviour, growth, fecundity and mortality (Marcogliese, 2004). Moreover, parasites may also regulate host population dynamics and influence community structure (Malan *et al.*, 1997; Marcogliese, 2004; Hatcher *et al.*, 2006; Vignon and Sasal, 2010). It is observed that the parasites act as indicators of changes in the ecosystem structure and function environment and thus, eventually reflecting the position of host in the food web. The study of ecological concepts of parasites provides an overall view of interrelationships between parasites, the host and the environment (Marcogliese, 2004). Parasitism

is very important biotic factor as in this association the host is under the influence of the parasite. The structure of metazoan parasite community of fish is receiving major attention of ecologists. Parasitic communities are a complex of individuals belonging to the different parasite species inhabiting a host (Cloutman, 1975). The study on seasonal dynamics of parasitism levels serves as a tool to understand broad aspects that determine the population biology of the host-parasite system (Chubb, 1982). Parasite communities generally lie somewhere along the interactive-to-isolationist continuum, i.e., from rich assemblages of species with high colonisation rates in which interspecific interactions play an important structuring role, to species-poor assemblages where interactions are unlikely. This framework has become one of the paradigms of parasite community ecology. Holmes and Price (1986) gave a theoretical consideration which shows that helminth communities span a continuum between interactive to isolationist. Vast majority of work has been focused on the metazoan parasitic communities in fishes including both ectoparasites and endoparasites (Thoney, 1991; Aho and Bush, 1993; Campos and Carbonell, 1994; Molloy *et al.*, 1995; Luque *et al.*, 1996a, b; Alves and Luque, 2001a, b, Aloo, 2002; Alves *et al.*, 2002; Guidelli *et al.*, 2003; Aguilar-aguilar and Salgado-Maldonado, 2006; Munoz *et al.*, 2007; Hernandez *et al.*, 2007). The present study aims to develop an analysis of community structure in the Mastacembelid fish, *Mastacembelus armatus*. *Mastacembelus armatus* Lacépède, 1800 is the largest spiny eel of the genus *Mastacembelus*. It is a popular indigenous aquarium fish and also an economically important food fish (Sugunan *et al.*, 2002; Tripathi, 2004). It is esteemed as highly proteinaceous food and attributed as tasty and medicinally important fish. They are commonly known as zigzag eel, spiny eel, leopard spiny eel and white-spotted spiny eel. It is also locally called as Pedda papera or papera or freshwater Baam or Bommidai and harbours a variety of metazoan parasitic fauna which includes monogeneans, digeneans, cestodes, nematodes, acanthocephalans, copepods and isopods. Taxonomic studies on metazoan parasites of *M. armatus* from various parts of the world includes those of Harshey (1937), Karve and Naik (1951), Gupta (1958), Jain (1958, 1959), Tripathi (1959), Kulkarni (1969), Verma (1973), Shinde and Jadhav (1980), Venkatanarsaiah (1981), Agarwal and kumar (1981), Agrawal and Singh (1982), Agrawal and Agarwal (1988), Agrawal and Sharma (1989), Desmukh and Shinde (1989), Hasnain (1992), Jadhav *et al.* (1990), Dubey *et al.* (1992), De and Dey (1992), Noopur *et al.* (1994), Wongsawad and Jadhav (1998), Kritsky *et al.* (2004), Jalali and Barzegar (2006) and Jalali *et al.* (2008). However, there are no studies on the populational and ecological features of metazoan parasites in these fishes. Hence, in this report, we analyse at the component and infracommunity levels a metazoan parasite community of *M. armatus* from the river Godavari, India.

MATERIALS AND METHODS

494 *Mastacembelus armatus* were procured from the river Godavari, Rajahmundry and local fish markets in and around the river and were brought to the laboratory for the examination of parasites from August 2005 to September 2007. Fishes were identified according to Jayaram (1981), Munro (1982) and Day (1994). Length, weight and sex of each fish individually were noted carefully and all organs were examined separately for the collection of parasites. Parasites were collected, enumerated and permanent slides were prepared by preserving the parasites in AFA (Alcohol, Formaldehyde and Acetic acid in 85:10:5) and stained with Alum carmine (Hiware *et al.*, 2003; Madhavi *et al.*, 2007). *M. armatus* measured 18-52 cm (mean = 36.42 ± 6.69 cm) in total length. The average total length of male (35.4 ± 6.59 cm, n = 221) and average length of female (37.32 ± 6.64 cm, n = 273) fishes in the sample were not significantly different (t = -1.0, p = 1.00).

The dominance frequencies of each parasite infracommunity were calculated by Berger-Parker index of dominance (Magurran, 1988). Pearson's correlation coefficient r was used as an indication of the relationship between the host's length and prevalence of parasites. The effect of host sex on abundance and prevalence of parasites was tested by applying chi-square test. Parasite species diversity was calculated using Shannon-Weiner index (H'). For each infracommunity, the evenness (Shannon-based evenness index, E) was calculated. Diversity H' and evenness (E) were calculated only for those fish with more than one parasitic species (diversity not defined for $K=1$).

RESULTS

Community ecology of *Mastacembelus armatus*: Twelve species of metazoan parasites (6 species of digenea, 2 cestodes, 1 monogenea, 1 copepoda, 1 nematoda and 1 acanthocephala) were collected from the host during the study period (Table 1). In *M. armatus*, the parasitic fauna is predominated by digeneans followed by cestodes, monogeneans, copepods and the remaining groups at negligible level. This is evidenced by the Berger-parker's dominance indices and mean total parasites (Table 2). 187 hosts (38%) of *M. armatus* showed infection with single parasitic group, 158 (32%) with two, 36 (7%) with three and a negligible number i.e., 2 (less than 1%) with four parasitic groups respectively (Table 3). No single fish has found infected with all the parasitic groups simultaneously. Of the six species of Digenea, two are larval stages, *Tetracotyle* sp. and *Ascocotyle* sp. and other four are adults, *Genarchopsis faruquis*, *Phyllodistomum tripathii*,

Table 1: Diversity parameters and distribution patterns of parasitic species of *Mastacembelus armatus*

Name of the parasite	Infected fishes	No. of parasites	Prevalence	Mean intensity	Mean abundance	Index of infection
<i>M. heteranchoratus</i>	74	469	15.10	4.3±1.60	0.95±1.80	0.32
<i>Genarchopsis faruquis</i>	105	391	21.40	2.9±1.83	0.79±1.40	0.30
<i>Phyllodistomum tripathii</i>	65	174	13.20	1.84±0.5	0.35±0.67	0.10
<i>Opcoelus mehrii</i>	133	616	26.40	3.7±2.08	1.24±2.03	0.50
<i>Allogeomtiotrema armati</i>	29	88	5.94	1.3±1.24	0.18±0.50	0.029
<i>Tetracotyle meatercaria</i>	230	31333	46.60	140.0±87.5	63.40±92.2	30.60
<i>Ascocotyle metacercaria</i>	64	257	13.02	3.30±1.1	0.52±1.25	0.11
<i>Circumnonchobothrium</i> sp.	184	536	37.67	2.50±1.22	1.10±1.45	0.57
Plerocercoid of <i>Senga</i> sp.	47	147	8.43	1.76±0.5	0.30±0.67	0.068
<i>Pallisentis colisai</i>	04	07	0.74	0.10±0.08	0.014±0.046	0.0011
<i>Neoergasilus indicus</i> n.sp.	41	110	8.13	1.10±0.36	0.22±0.45	0.056
<i>Camallanus unispiculus</i>	30	68	6.61	0.79±1.22	0.15±0.322	0.043
Name of the parasite	Range	D.I	Mean total parasite species	Location	Nature of infection	Nature of species
<i>M. heteranchoratus</i>	1-22	0.0137	0.949	Gills	Frequent	Satellite
<i>Genarchopsis faruquis</i>	1-15	0.0114	0.79	Intestine	Frequent	Satellite
<i>Phyllodistomum tripathii</i>	1-6	0.0050	0.35	Urinary bladder	Frequent	Satellite
<i>Opcoelus mehrii</i>	1-18	0.0180	1.246	Intestine	Frequent	Satellite
<i>Allogeomtiotrema armati</i>	1-11	0.0026	0.178	Intestine	Rare	Satellite
<i>Tetracotyle meatercaria</i>	2-558	0.916	63.42	Body cavity, major body organs, heart	Common	Secondary
<i>Ascocotyle metacercaria</i>	1-12	0.0075	0.520	Gills	Frequent	Satellite
<i>Circumnonchobothrium</i> sp.	1-13	0.0145	1.084	Intestine	Common	Secondary
Plerocercoid of <i>Senga</i> sp.	2-8	0.0043	0.29	Intestine	Rare	Satellite
<i>Pallisentis colisai</i>	1-3	0.0002	0.0141	Intestine	Sporadic	Satellite
<i>Neoergasilus indicus</i> n.sp.	1-6	0.0032	0.222	Gills	Rare	Satellite
<i>Camallanus unispiculus</i>	1-6	0.0019	0.137	Intestine	Rare	Satellite

Table 2: Parasitic groups, number of parasites obtained, dominance index and mean total parasites in *M. armatus*

Parasite group	No. of parasites	Dominance index	Mean total parasites
Monogeneans	469	0.0137	0.95
Digeneans	32859	0.960	66.52
Cestodes	683	0.020	1.38
Nematodes	68	0.002	0.14
Acanthocephalans	07	0.0002	0.01
Copepods	110	0.0032	0.22

Table 3: Frequency distribution of number of parasitic species per individual in *Mastacembelus armatus*

No. of parasitic groups	No. of infected fishes	% of frequency
1	187	37.85
2	158	31.98
3	36	7.28
4	2	0.404
5	0	0

Table 4: Frequency distribution of number of parasitic species per individual in *M. armatus*

No. of parasitic species	No. of infected fishes	% of frequency
1	87	17.60
2	112	22.60
3	91	18.42
4	56	11.30
5	30	6.07
6	7	1.40
7	0	0.00

Table 5: Diversity parameters of metazoan parasite communities of *M. armatus*

Host	Sample size	Mean No. of parasite species	Mean No. of parasite individuals	Shannon's diversity index (H')	Shannon-based evenness (E)
<i>Mastacembelus armatus</i>	494 (196)	2.20±0.42	84.30±106.8	0.71±0.47	0.93±0.59
		Range (2-4)	Range (2-567)	Range (0.021-1.75)	Range (0.031-2.54)

Allogomtiotrema armati and *Opecoelus mehrii*. The parasite fauna of the host is exclusively predominated by the larval stages of the strigeid trematode, *Tetracotyle* sp. with 31333 parasites and constitutes about 96% of the total parasites collected. Two species of cestodes, *Circumonchobothrium shindei* together with its plerocercoid stages and plerocercoid stage of *Senga* sp. dominates the parasite fauna next to *Tetracotyle* sp. Of the 494 fishes examined, 383 (78%) fish were frequently parasitized with one or more than one parasite species. A total of 34,196 (Range = 2-558) individual parasites was collected with a mean of 67 parasites/fish. Eighty seven (18%) hosts showed infection with single parasite species, 112 (23%), 9 (18%), 56 (11%), 30 (6%) and only 7 (1%) hosts showed multiple infections with 2, 3, 4, 5 and 6 parasite species respectively (Table 4). Mean parasite species richness is 2.20±0.42. The mean parasite diversity (Shannon's H' index) is 0.714±0.47 and the maximum diversity is 1.75. Shannon-based evenness index (E) has a mean of 0.927±0.597 (Table 5). *Tetracotyle* larvae showed the highest value of index of dispersion, followed by metacercariae of *Ascocotyle nana* and *Mastacembelocleidus heteranchoratus*

Table 6: Comparison of the mean (X), Variance (s^2) and Dispersion index (s^2/X) of parasite species in *M. armatus*

Name of the parasite	2005-2006 (n = 245)				2006-2007 (n = 249)			
	No. of parasites collected	Mean (X)	Variance (s^2)	Dispersion index (s^2/X)	No. of parasites collected	Mean (X)	Variance (s^2)	Dispersion index (s^2/X)
<i>Mastacembelocleidus heteranchoratus</i>	233	0.95	10.16	10.6	236	0.94	7.21	7.67
<i>Tetracotyle metacercaria</i>	14911	60.86	7714.04	126.75	16422	65.95	10972.1	166.37
Metacercaria of <i>Ascocotyle nana</i>	108	0.44	4.99	11.3	149	0.598	5.78	9.66
<i>Opegaster mehrii</i>	298	1.22	12.30	10.08	318	1.29	6.53	5.06
<i>Genarchopsis faruquis</i>	213	0.86	7.06	8.2	178	0.71	3.011	4.24
<i>Phyllodistomum tripathii</i>	81	0.33	0.88	2.66	93	0.37	0.99	2.67
<i>Circumonchobothrium</i> sp.	251	1.02	2.89	2.83	285	1.14	2.95	2.58

for both the cycles (Table 6). The calculated values of chi-square for both the years are much lower than the table values suggesting the fitting of Poisson distribution pattern for the parasitic species. Grabda-Kazubski *et al.* (1987) classified the nature of infection by parasite species into dominant (>70%), sub-dominant (50-70%), common (30-50%), frequent (10-30%), rare (4-10%) and sporadic (<4%) based on the percentage of prevalence (Table 1). In the present study, *Tetracotyle* sp. and *Circumonchobothrium shindei* show common infection with a prevalence ranging between 30-50%. Plerocercoid of *Senga*, *Neoergasilus indicus* and *Camallanus unispiculus* show rare infections (4-10%). *Pallisentis colisai* shows sporadic infection (<4%) whereas *M. heteranchoratus* and metacercaria *Ascocotyle nana* shows frequent infection. The abundance of parasitic species, range, prevalence, mean intensity, mean abundance and index of infection were calculated (Table 1). In the present investigation, based on the studies of Bush *et al.* (1997) it is observed that there is no core species in the host. Only secondary and satellite species were encountered. Two species, metacercaria *Tetracotyle* sp. and *C. shindei* occur in large numbers and occupy the position of Secondary species while the rest of the parasite species, monogenean-*M. heteranchoratus*, digeneans-metacercaria *A. nana*, *Genarchopsis faruquis*, *Phyllodistomum tripathii*, *Opecoelus mehrii*, *Allogomtiotrema armati*, copepod-*N. indicus*, nematode-*C. unispiculus*, cestode-Plerocercoid of *Senga* sp. and Acanthocephalan-*P. colisai* are satellite species (Table 1). Mean intensity and mean abundance are exceptionally very high for *Tetracotyle* sp., moderate for rest for the species and negligible in *P. colisai*. Berger-Parker's dominance index and mean of total parasite species was calculated for each species of parasite in the hosts (Table 1). *Tetracotyle* sp. presented the highest dominance value of 0.916 followed by the digenean, *O. mehrii* (0.018) and cestode, *C. shindei* (0.0145). Mean of total parasitic species of *Tetracotyle* sp. is 63.42, 1.246 for *O. mehrii* and 1.084 for *C. shindei*. The pairwise associations between the larval helminths were calculated exclusively since the parasitic fauna of the fish is predominated by larval helminths. The pairwise associations were calculated for *A. nana* and *Tetracotyle* sp. in *M. armatus* by using Jaccard's coefficient (r_j) and modified Sørensen's coefficient. Jaccard's coefficient of association was low for one pairwise combination in *M. armatus* suggests that the two species in a pair do not co-occur very often in the same fish individual (Table 7). However, Sørensen's coefficient for the pairwise combinations in the fishes showed negative coefficient values suggesting that the two species in a pair do not co-occur recurrently in the same fish (Table 8). Host size also serves as a factor in determining the burden of parasitic communities in a host. In the present study, *M. armatus* from 18 cm to 53 cm were collected and were categorized into 9 classes. The possible

Table 7: Matrix of pairwise associations between larval helminths using Jaccard's coefficient, r_j ; each fish sample treated separately

Host species	<i>M. armatus</i>	
	<i>Tetracotyle</i> sp.	<i>Ascocotyle</i> sp.
<i>Tetracotyle</i> sp.	-	0.168
<i>Ascocotyle</i> sp.	0.168	-

Table 8: Matrix of pairwise associations between larval helminths using modified Sorensen's coefficient, r_s ; each fish sample treated separately

Host species	<i>M. armatus</i>	
	<i>Tetracotyle</i> sp.	<i>Ascocotyle</i> sp.
<i>Tetracotyle</i> sp.	-	-0.676
<i>Ascocotyle</i> sp.	-0.676	-

Table 9: Correlation of parasite species richness, parasite species diversity and evenness in different size classes of *M. armatus*

Size class	Parasite species richness	Parasite species diversity (H')	Evenness (E)	Correlation coefficient (r)
18-21	2.50±0.70	0.81±1.00	0.77±0.88	0.411
22-25	2.00±0.00	1.00±0.01	1.45±0.016	
26-29	2.20±0.44	0.44±0.45	0.52±0.58	
30-33	2.14±0.36	0.75±0.46	0.98±0.58	
34-37	2.08±0.28	0.68±0.45	0.93±0.60	
38-41	2.20±0.40	0.79±0.50	1.02±0.64	
42-45	2.24±0.48	0.71±0.42	0.92±0.52	
46-49	2.44±0.62	0.72±0.54	0.89±0.71	
50-53	2.22±0.44	0.45±0.35	0.55±0.36	

interrelationship between the host size and parasite burden were carried out by Pearson's correlation tests. Pearson's coefficient was 0.411 for *M. armatus* which suggests that the total body size and total parasite burden is positively correlated. Overall parasite richness and parasite diversity are poorly correlated with the host size. However, 'r' values have also been calculated for individual species in relation to host body length (Table 9). Of the seven individual species most prevalent species (>10%) the cestode, *C. shindei* ($r = 0.461$), copepod, *N. indicus* ($r = 0.514$) and digeneans, *A. nana* ($r = 0.706$), *Tetracotyle* sp. ($r = 0.402$) and *O. mehrui* ($r = 0.334$) and the monogenean, *M. heteranchoratus* ($r = 0.638$) presented significant correlations between host total length and parasite abundance. Host sex is one of the biotic factors which play an important role in determining the parasitization in a host. Of the total sample of 494 *M. armatus*, 273 were females (55%) and 221 were males (45%). Among these fish, 225 females (46%) and 157 males (32%) were parasitized by one or more species of parasites. Based on a benchmark of 0.05 alpha, the estimated p-value of 0.00 suggests that there is no statistically significant association between the parasite abundance of males and females and that the host sex has no influence on parasitization. But the impact of host sex on individual parasite species were studied of which only the metacercaria of *Tetracotyle* sp. and digenean, *G. faruquis*, showed relatively higher values of prevalence and intensity of infection in females than in males (Table 10). To find the significant differences in parasite diversity of males and females, t-test was conducted. The estimated values of $t = -0.405$, $p < 0.686$ suggests that there is no statistically significant difference in the parasite diversity between males ($H' = 0.73 \pm 0.47$) and females ($H' = 0.702 \pm 0.474$).

Table 10: Diversity parameters of parasitic species in males and females of *M. armatus*

Host name	<i>Mastacembelus armatus</i>							
Parasite	N _{mi}	N _{fi}	P _m	P _f	MI _m	MI _f	MA _m	MA _f
Monogenea								
<i>M. heteranchoratus</i>	25	49	11.30	17.90	7.52	5.73	0.85	1.03
Digenea								
<i>Ascocotyle</i> sp. (M)	24	40	10.86	14.65	3.96	4.05	0.43	0.59
<i>Tetracotyle</i> sp. (M)	87	143	39.36	52.38	124.80	143.20	49.14	74.98
<i>Genarchopsis faruquis</i>	39	66	17.64	24.17	3.66	3.75	0.64	0.908
<i>Opegaster mastacembeli</i>	53	80	23.98	29.30	4.56	4.67	1.09	1.36
<i>Allogomtiotrema armati</i>	12	17	5.42	6.22	2.66	3.29	0.144	0.205
<i>Phyllodistomum tripathi</i>	28	37	12.66	13.55	2.50	2.81	0.32	0.38
Cestoda								
<i>Circumonchobothrium</i> sp.	80	104	36.20	38.10	2.90	2.89	1.06	1.10
Plerocercoid of <i>Senga</i> sp.	16	25	7.24	9.15	3.40	3.72	0.24	0.34
Nematoda								
<i>Camallanus unispiculus</i>	13	17	5.88	6.23	1.92	2.52	0.11	0.16
Acanthocephalans								
<i>Pallisentis colisai</i>	1	3	0.45	1.09	1.00	2.00	0.004	0.022
Copeoda								
<i>Neoergasilus indicus</i> n.sp.	15	26	6.78	9.52	2.60	2.73	0.176	0.26

N_m : No. of males examined; N_f : No. of females examined; N_{mi} : No. of males infected; N_{fi} : No. of females infected; P_m and P_f : Prevalence of males and females respectively; MI_m and MI_f : Mean intensity of males and females; MA_m and MA_f : Mean abundance of males and females respectively

DISCUSSION

Some patterns in the structure and composition of the community of metazoan parasites of *M. armatus* were detected: (1) dominance of endoparasitic larval stage; (2) positive correlation of parasitic burden and richness and (3) No influence of host sex on parasitization. Majority of freshwater fishes act as second intermediate hosts for many adult avian trematodes (Sukontason *et al.*, 1999; Wongsawad *et al.*, 2004; Kue-A-Pai and Wiwanitkit, 2005) and hence the metacercarial infection is found to be heavy in these fishes. The density of larval parasitic fauna is found to be higher than diversity of adult parasitic spectrum. *M. armatus* is also considered as good intermediate or paratenic hosts since the parasitic fauna of these fishes is predominated by larval helminths. Larval helminths use fish as intermediate hosts for the completion of their life-cycles in the definitive vertebrate hosts (Luque and Poulin, 2004). Many scientists focussed their studies on the adult helminths in the gastrointestinal tract of fish; few on ectoparasites since these endohelminths and ectoparasites form two distinct guilds of parasites inhabiting the same host and compete for the same resources. But larval helminths does not form an interactive guild as they appear less likely to compete for resources since they seek shelter in various organs and because many of them are in the form of cysts and do not actively feed on host nutrients or tissues. The statistical association of larval helminths among fish hosts have received little attention. In the present study community ecology of larval helminths along with their associations and adults is considered. However, *M. armatus* depicts a different scenario with high parasitization throughout the year which can be attributed to the non-random assemblages of larval parasites, which is an adaptive strategy of larval helminths to increase their probability of reaching their next host successively. Diet, feeding habits, vagility of host species are the main factors affecting the parasitic

community structure especially for the parasites transmitted to their final host (Sasal *et al.*, 1997). Also the classical studies on the impact of diet and feeding habits on the parasitic fauna (community structure) of the hosts were carried out by Pearson (1968), Kuperman (1973), Cannon (1977), Williams and Jones (1994), Luque *et al.* (1996a) and Johnson *et al.* (2004). In the present study impact of food and feeding habits of the host can be considered as one of the prime cause of larval parasitic abundance as *M. armatus* are carnivorous and their diet constitutes insects, crustaceans, molluscs, snails and worms which act as primary intermediate hosts for the most of the digeneans and cestodes. Many parasitic communities are known to occupy precise, predictable and limited locations (sites) within their hosts (Crompton, 1973; Holmes, 1973; Crompton and Nesheim, 1976). Holmes and Price (1986) suggested that the number of species of parasites that a host species supports widely varies from host to host which might be due to the biotic and abiotic factors. Various factors responsible for the parasite composition within a host is host size, phylogeny, composition of diet, complexity of gut and vagility. It is true in case of *M. armatus* as these fishes occur in shallow waters where the movement is limited. Kennedy and Bush (1994) are of the opinion that the examination of helminth community structure at the level of host's family and genus over a particular geographical area will improve our understanding of unpredictable helminth communities and their determinant factors. This statement is in total agreement with the present study. It is interesting to note that the 12 species of both ecto and endoparasites reported in the present study. Some of them have been recorded previously from other localities from these hosts. However, a few parasitic species are reported for the first time from this locality. Only copepod, *Neoergasilus indicus* is the new species report encountered from *M. armatus* in the present study (Vankara and Chikkam, 2010).

Kennedy *et al.* (1986) hypothesized freshwater fish parasitic communities to be less diverse when compared to their counterparts' marine fishes due to poor diversity of intermediate hosts in freshwater environment. The richly nutritive habitat and the healthy nature of fish might have declined the diversity and density of infection when compared to the marine fishes. In the present study, *M. armatus* also exhibited poor parasitic diversity and holds good to the hypothesis of Kennedy *et al.* (1986). Pairwise interspecific associations among larval helminth species in *M. armatus* were carried out using two different coefficients of association (Jaccard's and Sorensen's coefficients of association). Values of J_r and J_s coefficients of association for *Tetracotyle-Ascocotyle* pairwise associations in *M. armatus* suggests that two species in a pair do not co-occur very often in the same fish individuals. The present study is in accordance with the views of Poulin and Valtonen (2001) which suggests that assemblages of larval helminth parasites in fish are not random selections of locally available species, but rather non-random packets of larval parasites that travel together along common transmission routes. Few parasitic species are narrowly specific to one species of hosts which are called as 'monoxenic' or 'stenoadaptive' parasites whereas the parasites with wide specificity constitute 'polyxenic' or 'euryadaptive' species (Margolis *et al.*, 1982). In the present study, none of the parasitic species of *M. armatus* are monoxenic. At the level of parasite component community, the concept of core and satellite species concept plays a crucial role in explaining the species interactions (Holmes and Price, 1986; Esch *et al.*, 1990; Sousa, 1994). In the present study, metacercaria *Tetracotyle* sp. and adult *Circumonchobothrium shindei* occupy the position of secondary species but these parasites inhabit different niches. *C. shindei* in the intestine and *Tetracotyle* sp. inhabiting each and every organ, heart, liver, coelom cavity etc. Helminth communities may be characterized as being isolationists or interactive and they are recognized only when the species regularly co-occur (Holmes and Price, 1986). *Tetracotyle* sp. and

C. shindei share with other species of trematodes and cestodes, so can be called as interactive species. Shotter (1973), Bell and Burt (1991), Machado *et al.* (1994), Poulin (1995), Fiorillo and Font (1996) and Luque *et al.* (1996a, b) stated that helminth diversity and parasite community richness are more consistently correlated with host size. Cotgreave (1993), Blackburn and Lawton (1994) and Blackburn and Gatson (1997) documented relationship between parasite abundance and body size. Takemoto and Pavanelli (1994) and Machado *et al.* (1994) found that increase in size and age of fish resulted a significant increase in the levels of parasitism. Llewellyn (1962), Pennycuick (1971a, b), Shotter (1973), McVicar (1977), Hanek and Fernando (1978a, b), Fernandez (1985), Valtonen *et al.* (1990) and Roubal (1990) found higher levels of parasitism in hosts with intermediate lengths. Fish acquire the parasites in their youth phase which are then eliminated in the fish's adult phase which might be due to ageing of parasites or immunological resistance of fish. The parasitization in *M. armatus* is in harmony with the above studies as the prevalence of infection is low in small fish, abundant in medium size and then decreased in larger fishes. Host sex is one of the biotic factors which are important in studying host-parasite relationships. Effects of host sex on the levels of parasitism was studied by Thomas (1964) which might be due to the physiological and biological factors and behavioural difference between males and females which produce a small albeit consistent sexual trend in infection levels. Females are more likely to be heavily infected when compared to males which might be due to the stress caused during reproductive periods and lead into behavioural changes, thus making them more vulnerable to heavy parasitization. In contrast, Folstad and Karter (1992) and Poulin (1996) found more parasitization in males which might be due to high levels of testosterone which causes immunosuppression making them more susceptible to parasites than females. But there are also studies by Lawrence (1970), Pennycuick (1971a, b), Kennedy (1975), Muzzall (1980), Belghyti *et al.* (1994), Takemoto and Pavanelli (1994, 2000), Machado *et al.* (1994), Luque *et al.* (1996a) and Lizama *et al.* (2005) who found host sex not to be a significant factor in determining the infection rate of helminth parasites in host fishes and the present study is in harmony with the above studies. However, individual parasites, *C. shindei* and *G. faruquis* presented a significant impact in relation to sex as females showed relatively higher parasitization for these parasites. The present ecological studies suggest that the parasite communities of *M. armatus* are predictable, depauperate and non-interactive. Parasite diversity and richness can be attributed to its phylogeny, presence of intermediate hosts in the area. Abundance of parasite population is attributed to feeding capacity of the host which is governed by biotic and abiotic factors like temperature, water currents etc. The present study also holds good with the views of Holmes (1990) which suggests that fish as intermediate hosts have rich parasite fauna since they harbour both adults and larval helminths. The freshwater fish parasitic communities are less diverse than their marine counterparts (Holmes, 1990).

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REFERENCES

- Agarwal, G.P. and R. Kumar, 1981. *Gangatrema chauhani* n.g., n.sp. (Trematoda: Opecoelidae, Plagioporinae) from the intestine of a freshwater eel, *Mastacembelus armatus* (Lac.). Japanese J. Parasitol., 30: 9-13.

- Agrawal, N. and H.S. Singh, 1982. On three new monogenetic trematodes from freshwater fishes at District Lucknow. Reading Zool., 1: 31-35.
- Agrawal, S.C. and G.P. Agarwal, 1988. *Dactylostomum harishaii* sp.nov. from the intestine of a freshwater eel, *Mastacembelus armatus* (Lac.). Indian J. Helminthol., 40: 44-46.
- Agrawal, S.C. and S.K. Sharma, 1989. *Dactylostomum satpali* sp.nov. from the intestine of a freshwater eel, *Mastacembelus armatus* (Lac.). Indian J. Helminthol., 41: 108-112.
- Aguilar-Aguilar, R. and G. Salgado-Maldonado, 2006. Diversity of helminth parasites of freshwater fishes in two hydrological basins in Mexico: Helminths and the Mexican beta diversity hypothesis. Interciencia, 31: 484-490.
- Aho, J.M. and A.O. Bush, 1993. Community Richness in Parasites of some Freshwater Fishes from North America. In: Species Diversity in Ecological Communities: Historical and Geographical Perspectives, Ricklefs, R.E. and D. Schluter (Eds.). University of Chicago Press, Chicago, pp: 185-193.
- Aloo, P.A., 2002. A comparative study of helminth parasites from the fish *Tilapia zillii* and *Oreochromis leucostictus* in Lake Naivasha and Oloidien Bay, Kenya. J. Helminthol., 76: 95-102.
- Alves, D.R. and J.L. Luque, 2001a. Community ecology of the metazoan parasites of the white-croaker *Micropogonias furnieri* (Osteichthyes: Sciaenidae) from the coastal zone of State of Rio de Janeiro, Brazil. Mem Inst Oswaldo Cruz., 96: 145-153.
- Alves, D.R. and J.L. Luque, 2001b. Quantitative aspects of metazoan parasites infrapopulations of *Micropogonias furnieri* (Osteichthyes: Sciaenidae) from the coastal zone of Rio de Janeiro, Brazil. Parasitological Dia., 25: 30-35.
- Alves, D.R., J.L. Luque and A.R. Paraguassu, 2002. Community ecology of the metazoan parasites of Pink Cusk-eel, *Genypterus brasiliensis* (Osteichthyes: Ophidiidae), from the coastal zone of the State of Rio de Janeiro, Brazil. Memorias Instituto Oswaldo Cruz., 97: 683-689.
- Belghyti, D., O. Berrada-Rkhami, V. Boy, P. Aguesse and C. Gabrian, 1994. Population biology of two helminth parasites of flatfishes from the Atlantic coast of Morocco. J. Fish Biol., 44: 1005-1021.
- Bell, G. and A. Burt, 1991. The comparative biology of parasite species diversity: Internal helminths of freshwater fish. J. Anim. Ecol., 60: 1047-1063.
- Blackburn, T.M. and J.H. Lawton, 1994. Population abundance and body size in animal assemblages. Philosophical Trans. Biol. Sci., 343: 33-39.
- Blackburn, T.M. and K.J. Gatson, 1997. A critical assessment of the form of the interspecific relationship between abundance and body size in animals. J. Anim. Ecol., 66: 233-249.
- Bush, A.O., K.D. Lafferty, J.M. Lotz and W. Shostak, 1997. Parasitology meets ecology on its own terms: Margolis *et al.* revisited. J. Parasitol., 83: 575-583.
- Campos, A. and E. Carbonell, 1994. Parasite community diversity in two Mediterranean labrid fishes *Symphodus tinca* and *Labrus merula*. J. Fish Biol., 44: 409-413.
- Cannon, L.R.G., 1977. Some ecological relationships of larval ascaridoids from south-eastern Queensland marine fishes. Int. J. Parasitol., 7: 227-232.
- Chubb, J.C., 1982. Seasonal occurrence of helminths in freshwater fishes: Part IV. Adult *Cestoda*, *Nematoda* and *Acanthocephala*. Adv. Parsitol., 20: 1-292.
- Cloutman, D.G., 1975. Parasite community structure of largemouth bass, warmouth and bluegill in Lake FortSmith, Arkansas. Trans. Am. Fish. Soc., 104: 277-283.

- Cotgreave, P., 1993. The relationship between body size and population abundance in animals. Trends Ecol. Evol., 8: 244-248.
- Crompton, D.W., 1973. The sites occupied by same parasitic helminths in the alimentary tract of vertebrates. Biol. Rev., 48: 27-83.
- Crompton, D.W.T. and M.C. Nesheim, 1976. Host parasite relationships in the alimentary tract of domestic birds. Adv. Parasitol., 14: 95-194.
- Day, F., 1994. The Fishes of India, Being a Natural History of the Fishes Known to Inhabit the Seas and Freshwaters of India, Burma and Ceylon. Vol. 2, Today Tomorrow's Printers and Publishers, India, ISBN: 8186095004.
- De, N.C. and J. Dey, 1992. A new species of the genus *Goezia* Zeder, 1800 (Nematoda: Anisakidae) from the fish, *Mastacembelus armatus* (Lac.) from West Bengal, India. Syst. Parasitol., 22: 189-197.
- Desmukh, S.V. and G.B. Shinde, 1989. On a new species *Siluritaenia behairrnathi* sp.nov. (Cestoda: Proteocephalidae) from a freshwater fish, *Mastacembelus armatus* from Aurangabad, M.S. Indian J. Helminthol., 41: 75-77.
- Dubey, A., A.K. Gupta and S.M. Agarwal, 1992. Studies on monogenean parasites of freshwater fishes at Raipur X. Taxonomic discussion on the validity of the genus *Urocleidus* (Müller, 1934) Müller, 1936 (Ancyrocephalinae) and description of a new species from *Mastacembelus armatus* and *Macrognathus aculeatus* at Raipur. Indian J. Helminthol., 44: 116-122.
- Esch, G.W., A.W. Shostak, D.J. Marcogliese and T.M. Goater, 1990. Patterns and Processes in Helminth Parasite Communities: An Overview. In: Parasite Communities: Patterns and Processes, Esch, G.W., A.O. Bush and J.M. Aho (Eds.). Chapman and Hall, London, UK., pp: 1-20.
- Fernandez, B.J., 1985. Estudio parasitológico de *Merluccius australis* (Hutton, 1872) (Pisces: Merlucciidae): apectos sistematicos, estadisticos y zoogeograficos. Biol. Soc. Biol. Concepcion, Chile, 56: 31-41.
- Fiorillo, R.A. and W.F. Font, 1996. Helminth community structure of four species of *Lepomis* (Osteichthyes: Centrarchidae) from an oligohaline estuary in South Eastern Louisiana. J. Helminthol. Soc. Washington, 63: 24-30.
- Folstad, I. and A.J. Karter, 1992. Parasites, bright males and the immunocompetence handicap. Am. Naturalist, 139: 603-622.
- Grabda-Kazubski, B., B. Batur-Warsza-wska and T. Pojmanska, 1987. Dynamics of parasite infestations of fishes in Lakes Dgal Wielki and Warnaik in connection with introduction of phytophagous species. Acta Parasitol. Pol., 32: 1-28.
- Guidelli, G.M., A. Isaac, R.M. Takemoto and G.C. Pavanelli, 2003. Endoparasite infracommunities of *Hemisorubim platyrhynchos* (Valenciennes, 1840) (Pisces: Pimelodidae) of the Baía river, upper Parana river floodplain, Brazil: Specific composition and ecological aspects. Brazilian J. Biol., 63: 261-268.
- Gupta, S.P., 1958. Two new trematodes of the family *Allocreadiidae* from the freshwater fishes of U.P. Indian J. Helminthol., 8: 100-106.
- Hanek, G. and C.H. Fernando, 1978a. The role of season, habitat, host age and sex on gill parasites of *Lepomis gibbous* (L.). Can. J. Zool., 56: 1247-1250.
- Hanek, G. and C.H. Fernando, 1978b. The role of season, habitat, host age and sex on gill parasites of *Ambloplites rupestris* (Raf.). Can. J. Zool., 56: 1251-1253.

- Harshey, K.R., 1937. On two new trematodes of the genus *Opegaster* Ozaki, with a systematic discussion on the families *Opecoelidae* Ozaki, 1925 and *Coitocaecidae* Ozaki, 1928. Proc. Ind. Aca. Sci., 5: 64-75.
- Hasnain, M., 1992. On a new cestode, *Senga chauhani* sp. nov from a fish host, *Channa punctatus* from Jamshedpur. Indian J. Helminthol., 44: 123-127.
- Hatcher, M.J., J.T.A. Dick and A.M. Dunn, 2006. How parasites affect interactions between competitors and predators. Ecol. Lett., 9: 1253-1271.
- Hernandez, A.D., J.F. Bunnell and M.V. Sukhdeo, 2007. Composition and diversity patterns in metazoan parasite communities and anthropogenic disturbance in stream ecosystem. Parasitology, 134: 91-102.
- Hiware, C.J., B.V. Jadhav and A.D. Mohekar, 2003. Applied Parasitology: A Practical Manual. Mangaldeep Publications, India.
- Holmes, J.C., 1973. Site selection by parasitic helminths: Interspecific interactions, site segregation and their importance to the development of helminth communities. Can. J. Zool., 51: 333-347.
- Holmes, J.C. and P.W. Price, 1986. Communities of Parasite. In: Community Ecology: Patterns and Process, Kikkawa, J. and D.J. Anderson (Eds.). Blackwell Scientific Publication, Oxford, pp: 187-213.
- Holmes, J.C., 1990. Competition, contacts and other factors restricting niches of parasitic helminths. Ann. Parasitol. Hum. Comp., 65: 69-72.
- Jadhav, B.V., A.B. Gavhane and B.W. Sawarkar, 1990. On a new tapeworm from *Mastacembellus armatus* at Achlapur, District Amaravati. Indian J. Parasitol., 16: 189-191.
- Jain, S.L., 1958. Monogenea of freshwater fishes. VII. *Bifurcohaptor*, a new genus from the gill filaments of two fishes from Lucknow. J. Parasitol., 44: 388-394.
- Jain, S.L., 1959. Monogenea of Indian freshwater fishes VIII. *Sprostonia*, a new genus of freshwater Tetraonchinae with the descriptions of four new species, from the gills filaments of fishes, from Lucknow. Parasitology, 49: 153-168.
- Jalali, B. and M. Barzegar, 2006. Fish parasites of Zarivar Lake. J. Agric. Sci. Technol., 8: 47-58.
- Jalali, B., M. Barzegar and H. Nezamabadi, 2008. Parasitic fauna of the spiny eel, *Mastacembelus mastacembelus* banks et solander (Teleostei: Mastacembelidae) in Iran. Iranian J. Vet. Res., 9: 158-161.
- Jayaram, K.C., 1981. The Fresh Water Fishes of India, Pakistan, Bangladesh, Burma and Sri Lanka: A Handbook. 1st Edn., Zoological Survey of India, Calcutta, pp: 475.
- Johnson, M.W., P.A. Nelson and T.A. Dick, 2004. Structuring mechanisms of yellow perch (*Perca flavescens*) parasitic communities: Host age, diet and local factors. Can. J. Zool., 82: 1291-1301.
- Karve, J.N. and G.G. Naik, 1951. Some parasitic nematodes of fishes-II. J. Univ. Bombay Section B Biol. Sci., 19: 1-37.
- Kennedy, C.R., 1975. Dispersion of Parasites within a Host-Parasite System. In: Ecological Animal Parasitology, Kennedy, C.R. (Ed.). Blackwell Scientific Publications, Oxford, London.
- Kennedy, C.R., A.O. Bush and J.M. Aho, 1986. Patterns in helminth communities: Why are birds and fish different? Parasitology, 93: 205-215.
- Kennedy, C.R. and A.O. Bush, 1994. The relationship between pattern and scale in parasite communities: A stranger in a strange land. Parasitology, 109: 187-196.
- Kritsky, D.C., K.C. Pandey, N. Agrawal and S.M.A. Abdullah, 2004. Monogenoids from the gills of spiny eels (Teleostei: Mastacembelidae) in India and Iraq, proposal of *Mastacembelocleidus* gen. n. and status of Indian species of *Actinocleidus*, *Urocleidus* and *Haplocleidus* (Monogeneoidea: Dactylogyridae). Folia Parasitol., 51: 291-298.

- Kue-A-Pai, P. and V. Wiwanikit, 2005. Contamination of the *Gnathostoma* spp. Infective stage larvae in *Fluta alba* sold in Bangkok. Internet Trop. Med., 2: 3-3.
- Kulkarni, T., 1969. Studies on the monogenetic trematodes of fishes found in Hyderabad, Andhra Pradesh (India). Part-III. Riv. Parassitol., 30: 263-282.
- Kuperman, B.I., 1973. Tapeworms of the Genus *Triaenophorus*: Parasites of Fishes. Amerind Publishing Co. Pvt. Ltd., New Delhi.
- Lawrence, J.L., 1970. Effects of season, host age and sex on endohelminths of *Catostomi commersoni*. J. Parasitol., 56: 567-571.
- Lizama, M. De Los A.P., R.M. Takemoto and G.C. Pavanelli, 2005. Influence of host sex and age infracommunities of metazoan parasites of *Prochilodus lineatus* (Val, 1836) (Prochilodontidae) of the Upper Parana River Floodplain, Brazil. Parasite, 12: 299-304.
- Llewellyn, J., 1962. The life histories and population dynamics of monogenean gill parasites of *Trachurus trachurus* (L). J. Mar. Biol. Assoc, U.K., 42: 587-600.
- Luque, J.L., J.F.R. Amato and R.M. Takemoto, 1996a. Comparative analysis of the communities of metazoan parasites of *Orthopristis ruber* and *Haemulon steindachneri* (Osteichthyes: Haemulidae) from the Southeastern Brazilian littoral: I. Structure and influence of size and sex of the hosts. Revista Brasileira Biologia, 56: 279-292.
- Luque, J.L., J.F.R. Amato and R.M. Takemoto, 1996b. Comparative analysis of the communities of metazoan parasites of *Orthopristis ruber* and *Haemulon steindachneri* (Osteichthyes: Haemulidae) from the Southeastern Brazilian littoral: II. Diversity, interspecific associations and distribution of gastrointestinal parasites. Revista Brasileira Biologia, 56: 293-302.
- Luque, J.L. and R. Poulin, 2004. Use of fish as intermediate hosts by helminth parasites: A comparative analysis. Acta Parasitol., 49: 353-361.
- Machado, M.H., G.C. Pavanelli and R.M. Takemoto, 1994. Influence of hosts sex and size on endoparasitic infrapopulations of *Pseudoplatystoma corruscans* and *Schizodon borelli* (Osteichthyes) of the high Parana river, Brazil. Revista Brasileira Parasitologia Veterinaria, 3: 143-148.
- Madhavi, R., C. Vijayalakshmi and K. Shyamasundari, 2007. Collection, Staining and identification of Different Helminth Parasites: A Manual of the Workshop on Fish Parasites-Taxonomy Capacity Building. Andhra University Press, India.
- Magurran, A.E., 1988. Ecological Diversity and its Measurement. 1st Edn., Princeton University Press, Princeton, ISBN: 0691084858.
- Malan, F.S., I.G. Horak, V. de Vos and J.A. van Wyk, 1997. Wildlife parasites: Lessons for parasite control in livestock. Vet. Parasitol., 71: 137-153.
- Marcogliese, D.J., 2004. Parasites: Small players with crucial roles in the ecological theatre. Ecohealth, 1: 151-164.
- Margolis, L., G.W. Esch, J.C. Holmes, A.M. Kuris and G.A. Schad, 1982. The use of ecological terms in parasitology (Report of an ad hoc committee of the American Society of Parasitologists). J. Parasitol., 68: 131-133.
- McVicar, A.H., 1977. Intestinal helminth parasites of ray, *Raja naevus* in British waters. J. Helminthol., 51: 11-21.
- Molloy, S., C. Holland and R. Poole, 1995. Metazoan parasite community structure in brown trout from two lakes in western Ireland. J. Helminthol., 69: 237-242.
- Munoz, G., A.S. Grutter and T.H. Cribb, 2007. Structure of the parasite communities of a coral reef fish assemblage (Labridae): Testing ecological and phylogenetic host factors. Int. J. Parasitol., 93: 17-30.

- Munro, I.S.R., 1982. The Marine and Freshwater Fishes of Ceylon. Soni Reprints Agency, Delhi, India.
- Muzzall, P.M., 1980. Population biology and host-parasite relationships of *Triganodistomum attenuatum* (Trematode: Lissorchiidae) infecting the white sucker, *Catostomus commersoni* (Lacepede). J. Parasitol., 66: 293-298.
- Noopur M., A.K. Srivastav and D. Rani, 1994. Piscian cestode of Bundelkhand region of Uttar Pradesh, *Senga jhansiensis* n.sp. (Cestoda: Ptychobothriidae) from *Mastacembelus armatus* (Lacepede). U.P J. Zool., 14: 94-96.
- Pearson, J.C., 1968. Observations on the morphology and life-cycle of *Paucivitellosus fragilis* Coil, Reid and Kuntz, 1965 (Trematoda: Bivesiculidae). Parasitology, 58: 769-788.
- Pennycuik, L., 1971a. Difference in the parasite infection in three spined stickle backs (*Gastersteus aculeatus*) of different sex, age and size. Parasitology, 63: 407-408.
- Pennycuik, L., 1971b. Seasonal variation in the parasite infection in a population of three-spined Sticklebacks, *Gastersteus aculeatus*. Parasitology, 63: 373-388.
- Poulin, R. and E.T. Valtonen, 2001. Nested assemblages resulting from host size variation. The case of endo parasite communities in fish hosts. Int. J. Parasitol., 31: 1194-1204.
- Poulin, R., 1995. Phylogeny, ecology and richness of parasite communities in vertebrates. Ecol. Monographs, 65: 283-302.
- Poulin, R., 1996. Sexual inequalities in helminth infections: A cost of being male? Am. Nat., 147: 287-295.
- Roubal, F.R., 1990. Seasonal changes in ectoparasite infection of juvenile yellowfin bream, *Acanthopagrus australis* (Günther) (Pisces: Sparidae), from a small estuary in Northern New South Wales. Aust. J. Mar. Freshwater Res., 41: 411-427.
- Sasal, P., S. Morand and J.F. Guegan, 1997. Determinants of parasite species richness in Mediterranean marine fish. Mar. Ecol. Prog. Ser., 149: 61-71.
- Shinde, G.B. and B.V. Jadhav, 1980. A new tapeworm *Senga godavari* n.sp. from *Mastacembelus armatus* at Aurangabad, India. Biology, 2: 46-48.
- Shotter, R.A., 1973. Changes in the parasite fauna of whiting *Odontogadus merlangus* L. with age and sex of the host, season and from different areas in the vicinity of the isle of man. J. Fish. Biol., 5: 559-573.
- Sousa, W.P., 1994. Patterns and processes in communities of helminth parasites. Trends Ecol. Evol., 9: 52-57.
- Sugunan, V.V., K. Mitra and G.K. Vinci, 2002. Ornamental fishes of West Bengal. Classic Printers, Kolkata, India.
- Sukontason, K., S. Piangjai, Y. Muangyimpong, K. Sukontason, R. Methanitikorn and U. Chaithong, 1999. Prevalence of trematode metacercariae in cyprinoid fish of Ban Pao district, Chiang Mai province northern Thailand. Southeast. Asian J Trop Med Public Health, 30: 365-370.
- Takemoto, R.M. and G.C. Pavanelli, 1994. Ecological aspects of proteocephalidean cestodes parasites of *paulicea leutkeni* (Steindachner) (Osteichthyes: Pimelodidae) from the Parana river, Parana, Brazil. Revista Unimar, 16: 17-26.
- Takemoto, R.M. and G.C. Pavanelli, 2000. Aspects of ecology of proteocephalid cestodes parasites of *Sorubim lima* (Pimelodidae) of the upper Parana river, Brazil: Structure and influence of hosts size and sex. Brazilian J. Biol., 60: 577-584.

- Thomas, J.D., 1964. A comparison between the helminth burdens of male and female brown trout, *Salmo trutta* (L.) from a natural population in River Teify, West Wales. Parasitology, 54: 263-272.
- Thoney, D.A., 1991. Population dynamics and community analysis of the parasite fauna of Juvenile spot, *Leiostomus xanthurus* Lac. and Atlantic croaker, *Micropogonias undulatus* Linn. (Sciaenidae) in two estuaries along the middle Atlantic coast of the United States. J. Fish. Biol., 39: 515-534.
- Tripathi, Y.R., 1959. Monogenetic trematodes from fishes of India. Ind. J. Helm., 9: 1-149.
- Tripathi, S.D., 2004. Ornamental fishes: Breeding, culture and trade. Proceedings of the National Seminar on Prospects of Ornamental Fish Breeding and Culture in Eastern and Northeastern India (NSPOFBCENT'04), Central Institute of Fisheries Education, Kolkata, India, pp: 17-42.
- Valtonen, E.T., M. Prost and R. Rakhonen, 1990. Seasonality of two gill monogeneans from two freshwater fish from an oligotrophic lake in Lake in Northeast Finland. Int. J. Parasitol., 20: 101-107.
- Vankara, A.P. and V. Chikkam, 2010. A new ergasilid copepod parasitic on the freshwater spiny eel, *Mastacembelus armatus* Lacepede, 1800 from river Godavari, India. Crustaceana, 83: 425-434.
- Venkatanarsaiah, J., 1981. New monogenetic trematodes of the genus *Dactylogyrus* diesing, 1850 from freshwater fishes of Andhra Pradesh, India. J. Anim. Morphol. Physiol., 28: 15-20.
- Verma, S.L., 1973. Helminth parasites of freshwater fishes. Part XI. On a new trematode, *Stomachicola mastacembeli* n.sp. from the intestine of a freshwater fish, *Mastacembelus armatus*. Zool. Anzeiger., 190: 167-170.
- Vignon, M. and P. Sasal, 2010. Fish introduction and parasites in marine ecosystems: A need for information. Environ. Biol. Fish., 87: 1-8.
- Williams, H. and A. Jones, 1994. Parasitic Worms of Fish. 1st Edn., Taylor and Francis, Bristol, UK, pp: 539.
- Wongsawad, C. and B.V. Jadhav, 1998. *Circumonchobothrium baimaii* n. sp. (Cestoda: Pseudophyllidea) from a freshwater fish, Maesa Stream, Chaing Mai, Thailand. Riv. Parassitol., 25: 291-294.
- Wongsawad, C., A. Rojanapaibul, P. Wongsawad, A. Rojanapaibul and T. Marayong *et al.*, 2004. Helminths of vertebrates in Mae Sa Stream, Chiang Mai, Thailand. Southeast Asian J. Trop. Med. Public Health, 35: 140-146.